1.2. What You Will Learn

ML Bootcamp - Syllabus

1. Course Introduction

Course roadmap and learning outcomes

2. Machine Learning Basics

Fundamental ML concepts and Machine Learning types

3. Python Environment Setup

Setting up tools required for ML development

4. Python Programming

Python essentials for Machine Learning.

5. Core Data Librariesfor ML

Working with data using Python libraries

6. Math Foundation for ML

Math intuition behind Machine Learning

7. Data Collection

Sourcing and loading ML datasets from different sources

8. Exploratory Data Analysis & Visualization

Understanding data through analysis and plots

Section 9 to 17 - Next Slide...

ML Bootcamp - Syllabus

9. Data Preprocessing

Cleaning and preparing data for modeling

10. Classification

Classification models & evaluation

11. Regression

Regression models & evaluation

12. Model training & Validation

Training models and evaluating performance

13. Model Optimization & Selection

Improving models and choosing the best one

14. Unsupervised Learning

Finding patterns without labeled data

15. Machine Learning Projects

Applying ML concepts to real problems

16. ML Model Deployment

Deploying models for real-world use

17. Capstone Projects - End-to-End

Building complete end-to-end ML applications

Learning Outcome

ML Fundamentals

Build strong intuition behind machine learning concepts, workflows, and decisions

Hands-on Practice

Learn by coding real examples, experiments, and projects step by step.

End-to-End ML Projects

Design and deploy end-to-end machine learning applications for real-world use.